Online algoritmer

torsdag den 28. november 2024 10.17

competitive Analysis:

OPT: optimal offline algorithm

Offline: hand (alt) data for computing

ALG: optimal online algorithm

online: der mangler information

start — online / offline → $\frac{ALG(I)}{OPT(I)} \leq c.$

c: bedste (mindste)

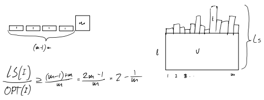
Skirental

OPT = købt dag 10

d = days to stay

if $d < 10$ rent every day

else $d \geq 10$ buy



$$\frac{LS(I)}{OPT(I)} \geq \frac{(m-1)+m}{m} = \frac{2m-1}{m} = 2 - \frac{1}{m}$$

t: sidste job definere makespan

$$Ls = l + t$$
$$\leq \frac{T-t}{m} \quad \text{fordi } l = \frac{V}{m} \text{ og } V \leq T - t$$
$$= \frac{T}{m} + \left(1 - \frac{1}{m}\right)t$$
$$\leq OPT + \left(1 - \frac{1}{m}\right)OPT$$
$$= (2 - \frac{1}{m}$$

Online problem

input arrives one request at a time

for each request, we have to make a decision

typically minimize costs...

Machine scheduling

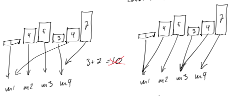
$m \geq 1$ machines.

n jobs of varying sizes arriving one at a time to be assigned to a machine.

goal: minimize makespan

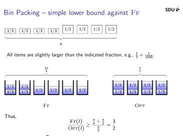
Algorithm List Scheduling (Ls): place jobs on least loaded machines

At Best $\left(2 - \frac{1}{m}\right)$



for any ALG and any I: $\frac{ALG(I)}{OPT(I)} \geq 1$

Bin Packing


Bin Packing – simple lower bound against FF

SDU

1/3 1/3 1/3 1/3 1/2 1/2 1/2 1/2

n

All items are slightly larger than the indicated fraction, e.g., $\frac{1}{3} + \frac{1}{1000}$.

$\frac{3n}{4}$ FF $\frac{n}{2}$ OPT

Thus,

$$\frac{FF(I)}{OPT(I)} \geq \frac{\frac{n}{4} + \frac{n}{2}}{\frac{n}{2}} = \frac{3}{2}$$

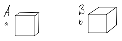
FF is 1,7 competitive [Hard]

dvs cr is at most 1.7

FF's cr is at least 1.5

FF interval is 1.5.. 1.7

jurel i boks



$$\frac{a+b}{\max[a,b]} = 2$$

juvel	forv. C.R.
A	$\frac{p \cdot a + (1-p)(a+b)}{a}$
B	$\frac{p(a+b) + (1-p) \cdot b}{b}$

$$\frac{p \cdot a + (1-p)(a+b)}{a} = \frac{p(a+b) + (1-p) \cdot b}{b}$$

$$pab + ab + b^2 - pab - pb^2 = pa^2 + pab + ab - pab$$

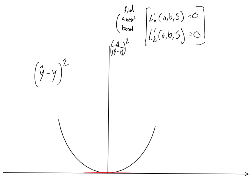
$$b^2 - pb^2 = pa^2$$

$$p = \frac{b^2}{a^2+b^2}$$

$$\frac{\frac{b^2}{a^2+b^2} \cdot a + \left(1 - \frac{b^2}{a^2+b^2}\right)(a+b)}{a}$$

$$\frac{b^2}{a^2+b^2} + \frac{1}{a}\left(\frac{a^2}{a^2+b^2}\right)(a+b)$$

$$\boxed{\frac{b^2 + a^2 + ab}{a^2+b^2}} \quad \text{if } a = b \quad cr = \frac{3}{2}$$



find (a best, b best): $\left[\begin{array}{l} L'_a(a,b,S) = 0 \\ L'_b(a,b,S) = 0 \end{array}\right]$

$$0 = \left(\sum_{i=1}^{3} (ax_i + b - y_i)^2\right)'_b$$
$$= \sum_{i=1}^{3} \left((ax_i + b - y_i)^2\right)'_b$$
$$= \sum_{i=1}^{3} (ax_i + b - y_i)$$
$$= 3b - \sum_{i=1}^{3} (y_i - ax_i)$$

$h = f + g$

$h' = f' + g'$

$f(g) = f'(g) \cdot g'$

$$L'_\theta(\theta, S) = \frac{2}{m} \sum_{i=1}^{m} (f(x_i;\theta) - y_i)(f'(x_i;\theta))$$

Learning rate $\alpha < 0$

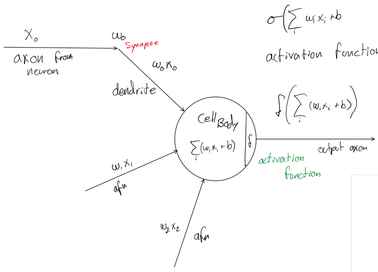
use default $\alpha = 0.001$

$\theta \leftarrow \theta - \alpha \cdot L(\theta, S)$

$$\theta \leftarrow \theta - \frac{2\alpha}{m} \sum_{i=1}^{m} (f(x_i;\theta) - y_i)(f'(x_i))$$

$\sigma\left(\sum_i w_i x_i + b\right.$

activation function



Clustering

Measure of compactness

$$TD^2(C) = \sum_{p \in C} \|p - \mu_C\|_2^2$$

$$TD^2(C_1, C_2 \ldots C_k) = \sum_{i=1}^{k} TD^2(C_i)$$

Loss function

$$L(a,b,S) = \sum_{i=1}^{3} \frac{1}{3}(ax_i + b - y_i)^2$$

0 1 2 3

0 1 0 1